Nourish SLM

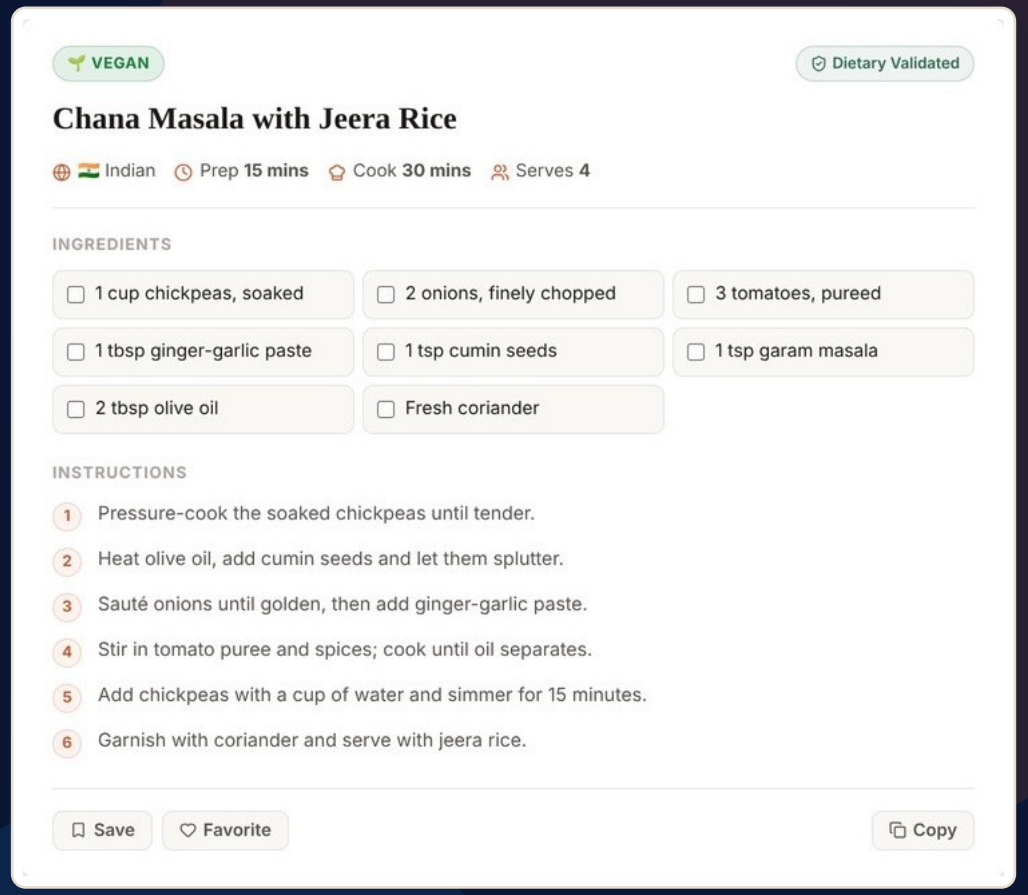
AI-Powered Recipe Assistant

A personalized food and diet assistant using a fine-tuned small language model

Cook what you have. Eat the way you live.

KM Sreenidhi
CSE (AI & ML), Woxsen University

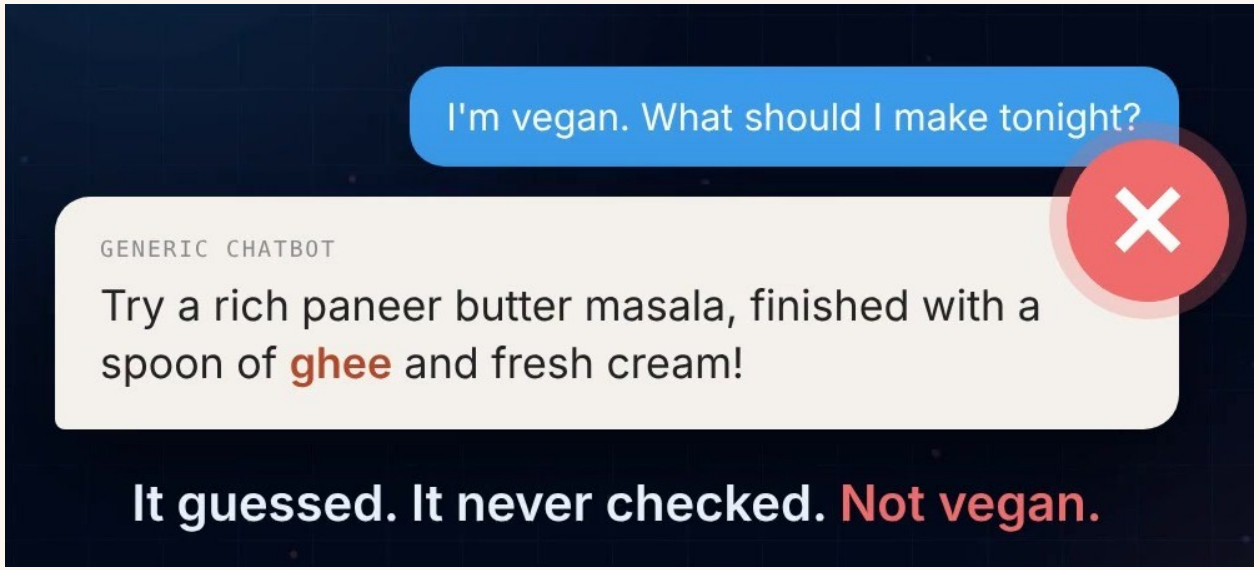
BUILT FOR VELTRIS
INNOVATE | ACCELERATE | TRANSFORM





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Chatbots guess. Nobody checks the diet.



A generic chatbot, asked for a vegan dinner

Never checks diet or allergies
Ghee for a vegan, peanuts for anyone

No nutrition facts
Calories and protein are a guess

No plan for your goals
Nothing built around training

Costly, not private
Paid cloud APIs see your data

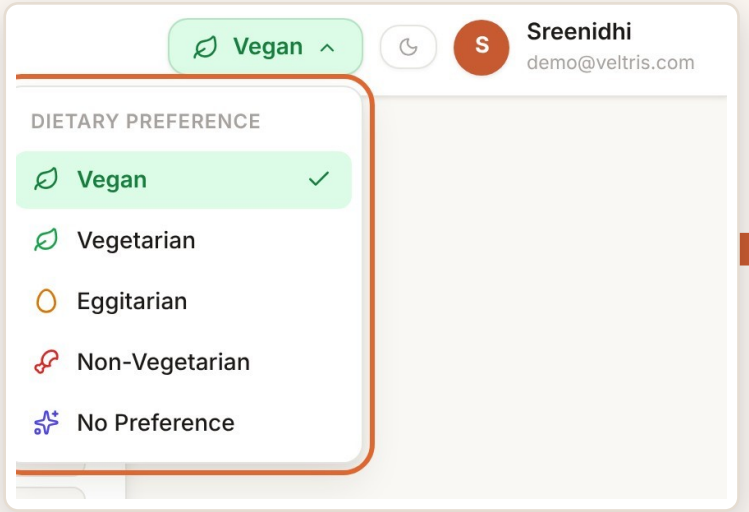
WHAT WE SET OUT TO BUILD

- Fine-tune Qwen3-4B
- Check diet + allergies
- Nutrition per serving
- Plans for 24 goals
- Beat the base model
- Run fully local

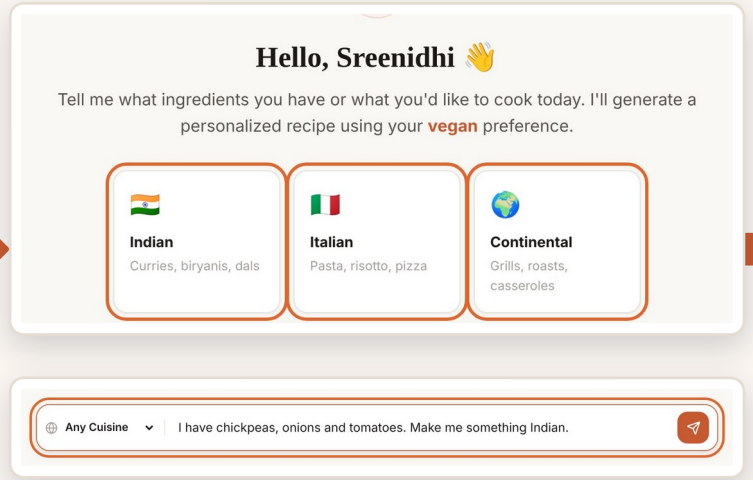
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Your kitchen in. A recipe that fits you out.

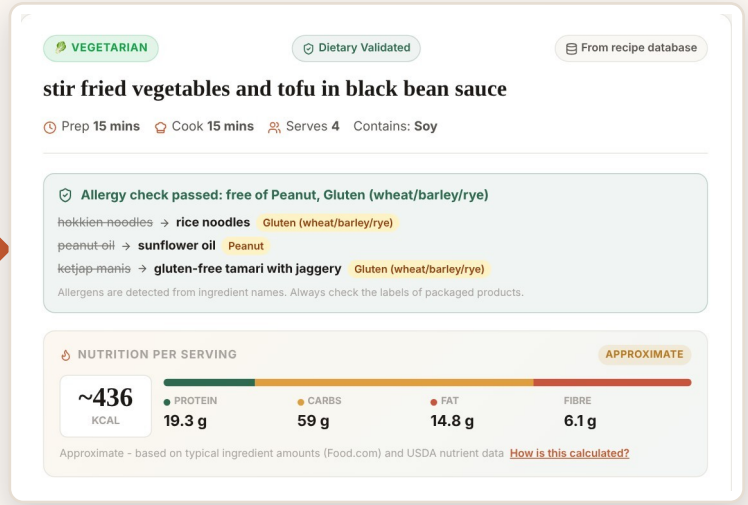
1 Pick diet + allergies



2 Say what you have



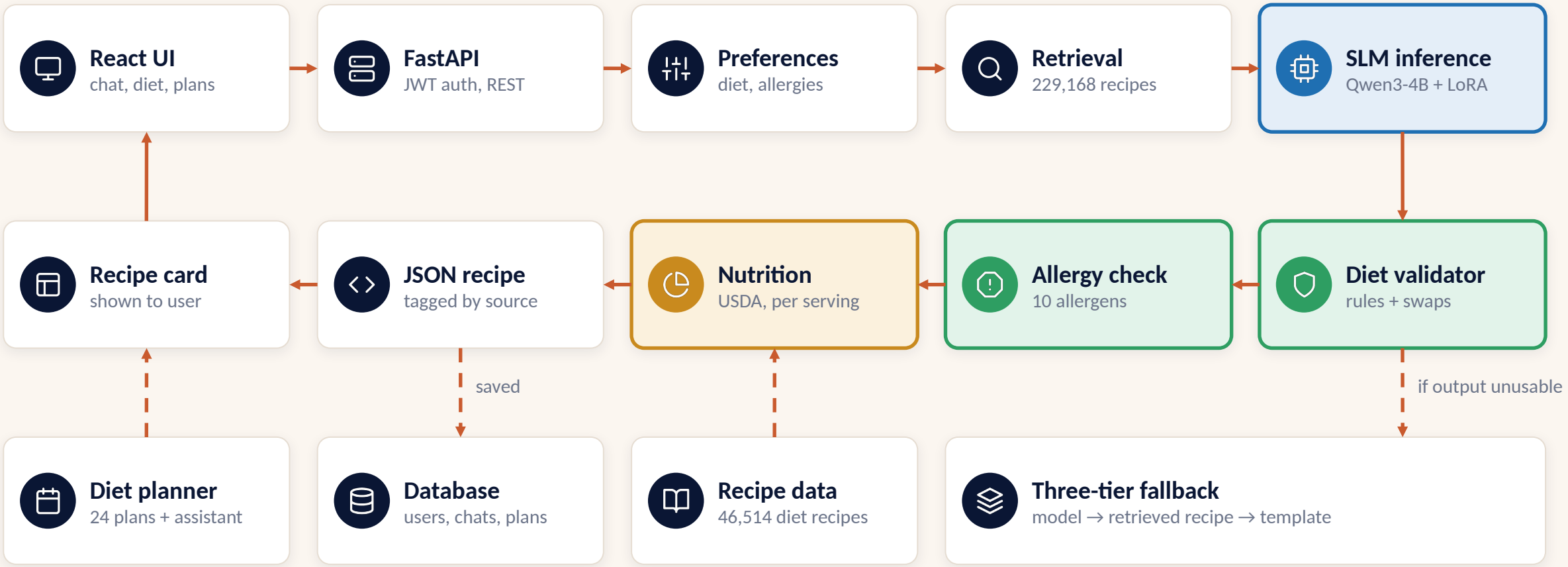
3 Get a checked recipe



- **Fine-tuned SLM**
- **Diet + allergy checks**
- **Nutrition per serving**
- **24 diet plans**

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The model writes. The validator decides.



Nothing reaches the user unchecked

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229,168 recipes taught Qwen3-4B to cook

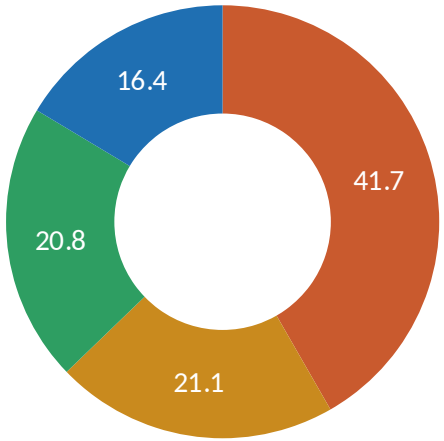
229,168
Food.com recipes, cleaned

6,000
balanced training examples

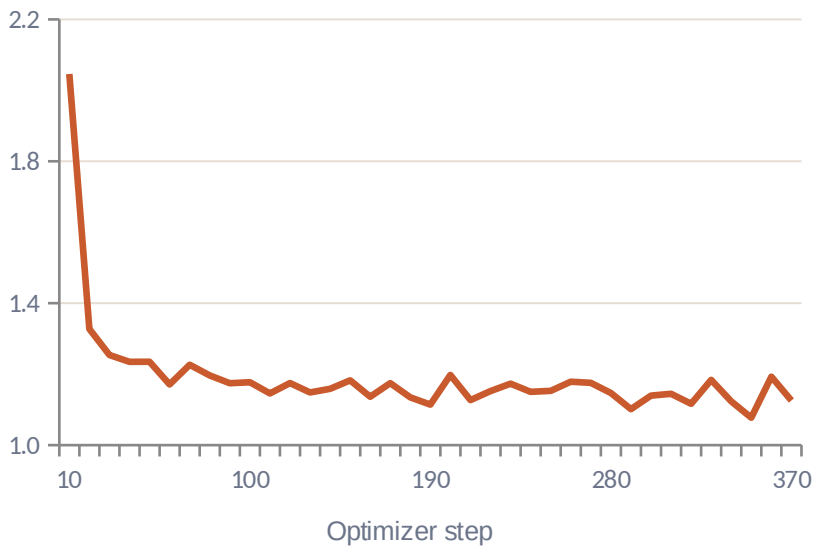
55 min
375 steps on one RTX 4070

132 MB
LoRA adapter on Qwen3-4B

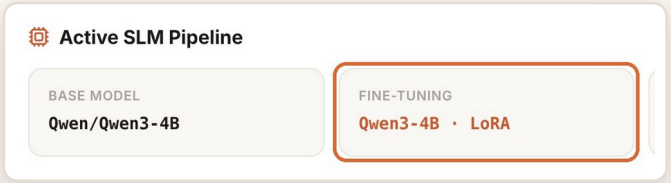
Diet labels (%)



Training loss falls fast



Live in the app



QLoRA recipe

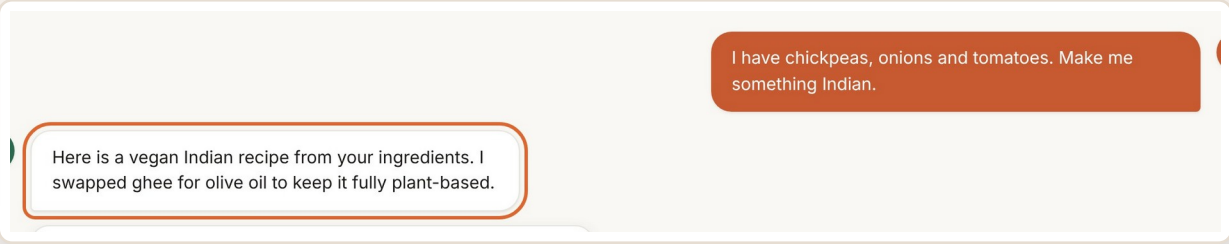
- 4-bit base
- r 16 · α 32
- all 7 projections
- lr 2e-4
- 1 epoch
- batch 16

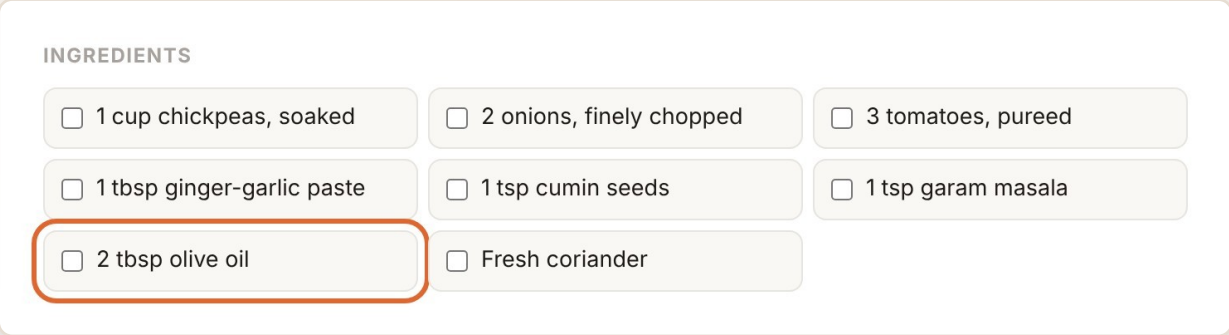
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Ghee in a vegan dish? Caught and swapped.

Model suggests → **ghee** dairy × vegan → **olive oil** plant ✓

What the user sees





Diet	Meat	Egg	Dairy	Plant
Vegan	✗	✗	✗	✓
Vegetarian	✗	✗	✓	✓
Eggitarian	✗	✓	✓	✓
Non-veg	✓	✓	✓	✓

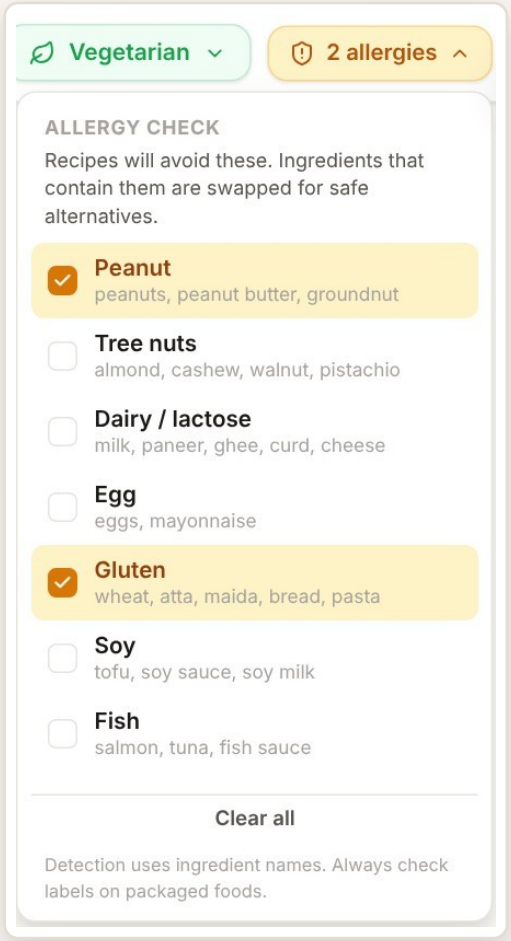
50 / 50
test cases pass

0.1 ms
per safety check



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Tick your allergies. Every recipe is made safe.



What the user sees

Here is your personalized Vegetarian recipe! Heads up: what you typed contains peanut, gluten (wheat/barley/rye), which is on your allergy list, so I replaced it.

Allergy check passed: free of Peanut, Gluten (wheat/barley/rye)

~~hokkien noodles~~ → **rice noodles** Gluten (wheat/barley/rye)

~~peanut oil~~ → **sunflower oil** Peanut

~~ketjap manis~~ → **gluten-free tamari with jaggery** Gluten (wheat/barley/rye)

Allergens are detected from ingredient names. Always check the labels of packaged products.

Example swaps

~~peanut oil~~	sunflower oil	Peanut
~~hokkien noodles~~	rice noodles	Gluten
~~paneer~~	firm tofu (or chickpeas)	Dairy
~~almond milk~~	oat milk	Tree nuts

10 allergens checked on every recipe

- Peanut
- Tree nuts
- Dairy
- Egg
- Gluten
- Soy
- Fish
- Shellfish
- Sesame
- Mustard

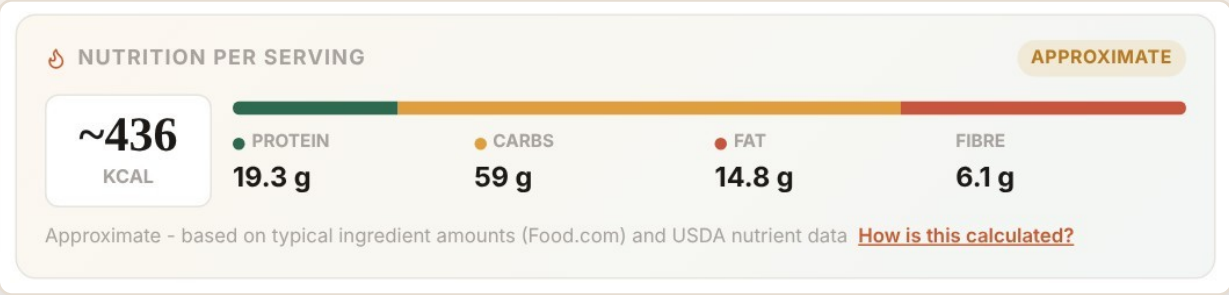
Swaps never add another of your allergens, and the steps are rewritten to match.

Anything that cannot be swapped gets a clear warning.

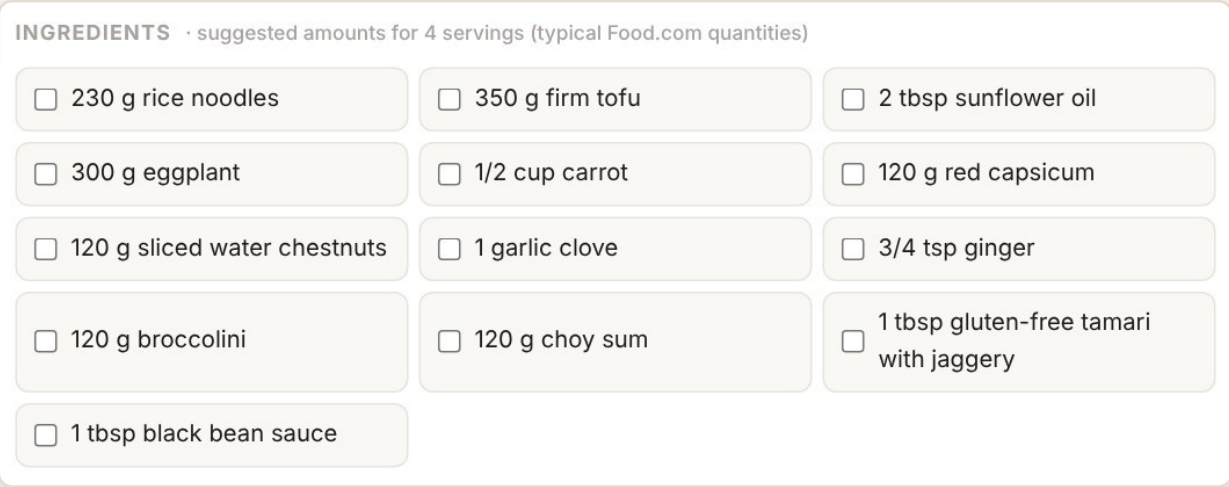
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Calories and macros on every recipe card

Per serving, on the recipe card



With suggested amounts for every ingredient



Dataset recipes

Published Food.com nutrition per serving, labelled Recipe data

Model-written recipes

The model lists ingredients without amounts, so:

1. Typical amount per ingredient: the median real Food.com recipes use
2. Nutrient values from USDA FoodData Central
3. Divided by servings and labelled Approximate

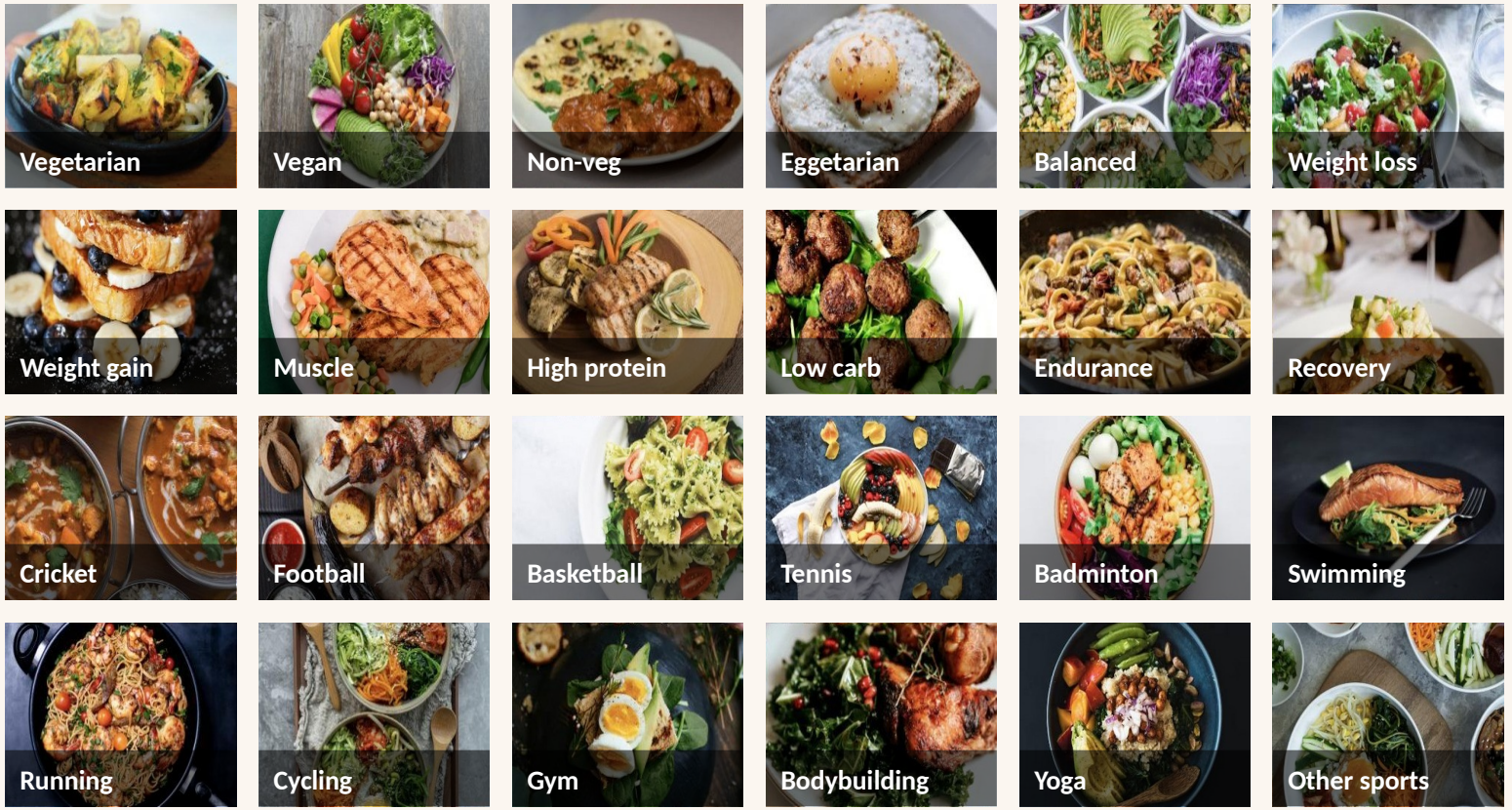
5
nutrients per serving

1,071
typical amounts

7,805
USDA foods

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24 diet plans, built from 46,514 real recipes



5 diet preferences

7 fitness goals

12 sports

46,514

plan-ready recipes with photos and nutrition

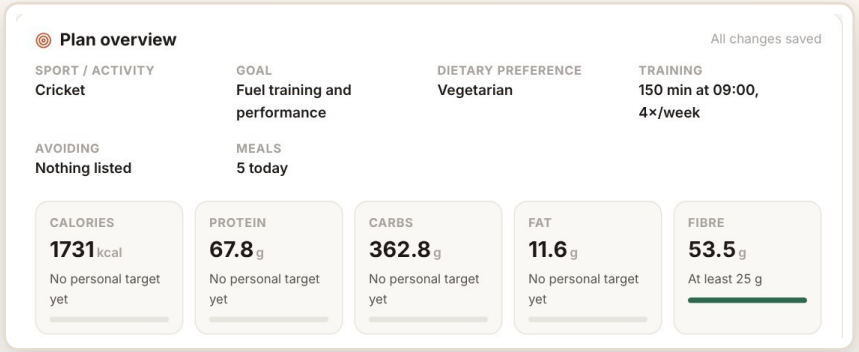
Targets from your own details

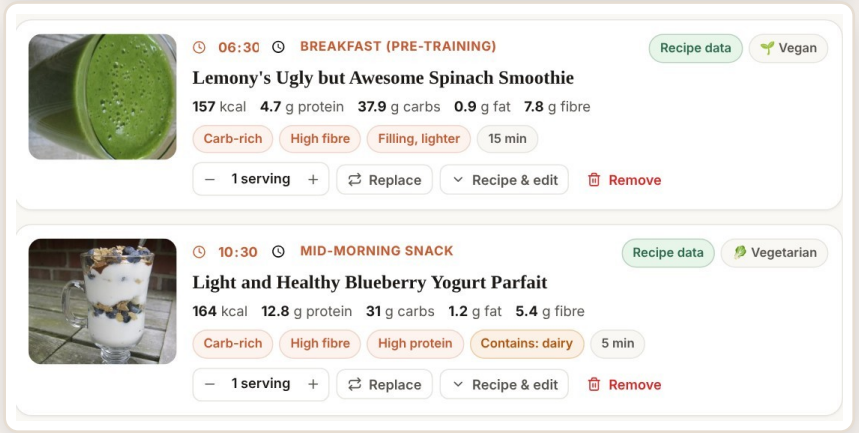
- Energy: Mifflin-St Jeor × activity level
- Protein 1.2–2.2 g/kg by goal (ACSM, ISSN)
- Carbs by training load (ACSM 2016)
- Meals timed around your training

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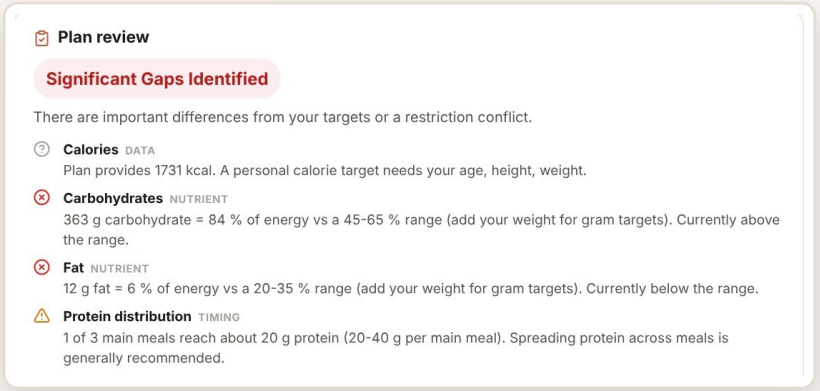
A day of meals, checked, with its own assistant

1 A full day of meals



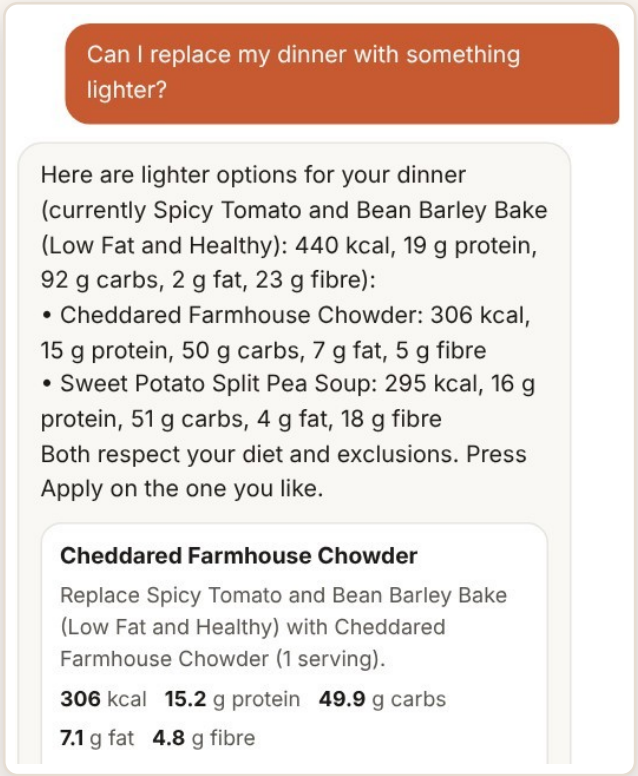


2 Check My Diet



- Restriction: a diet, allergy or dislike conflict
- Nutrient gap: calculated against your targets
- Data and timing: pre- and post-training meals

3 Plan assistant



Changes are proposals: the plan updates only when you press Apply

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An open-source stack, end to end

Frontend	React 19	TypeScript	Vite	Axios		
Backend	Python 3.12	FastAPI	SQLAlchemy	Pydantic	JWT + bcrypt	
AI / ML	Qwen3-4B	PyTorch	Transformers	LoRA / QLoRA	Pandas	Jupyter
Data	SQLite	PostgreSQL	Food.com	USDA FDC		
Deploy	Apple MPS	NVIDIA CUDA	Docker	Nginx	GitHub	

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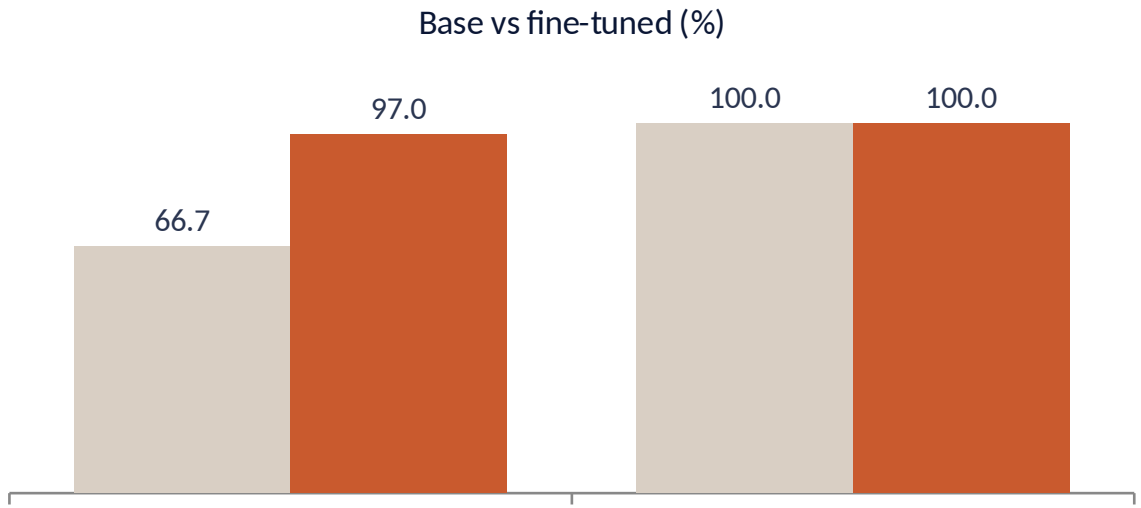
62% lower error. 97% complete recipes.

−62%
test loss (2.895 → 1.106)

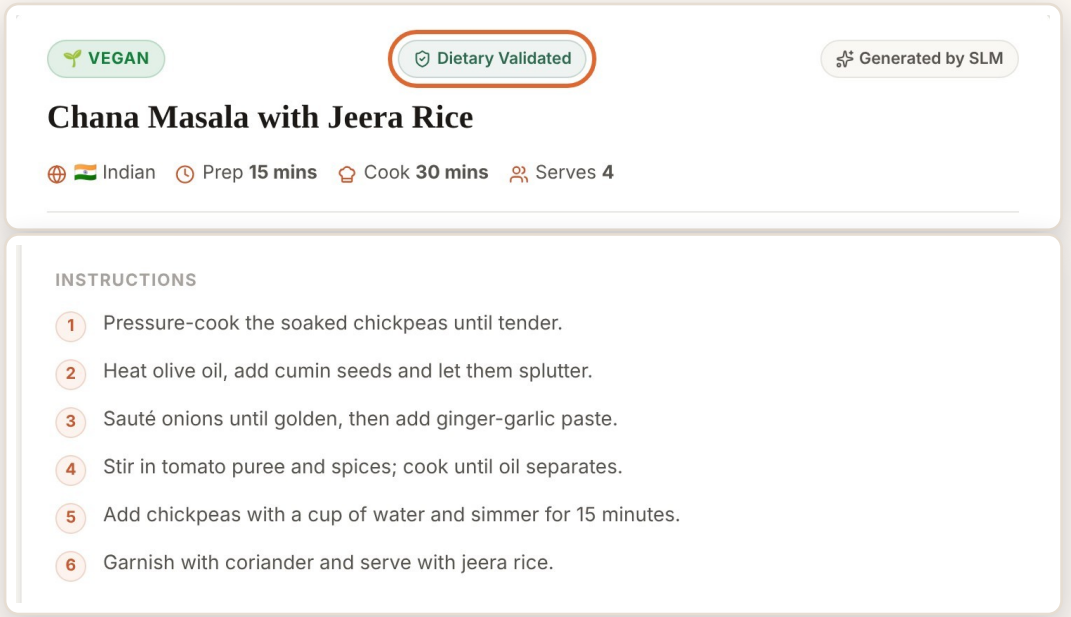
6×
lower perplexity (18.1 → 3.0)

97%
complete recipes (was 66.7%)

100%
diet compliance, kept

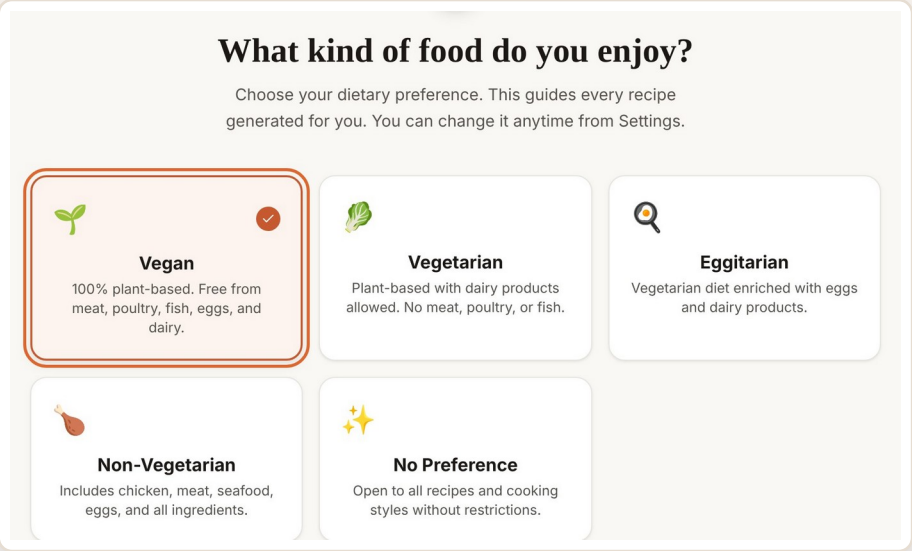


97% of answers look like this

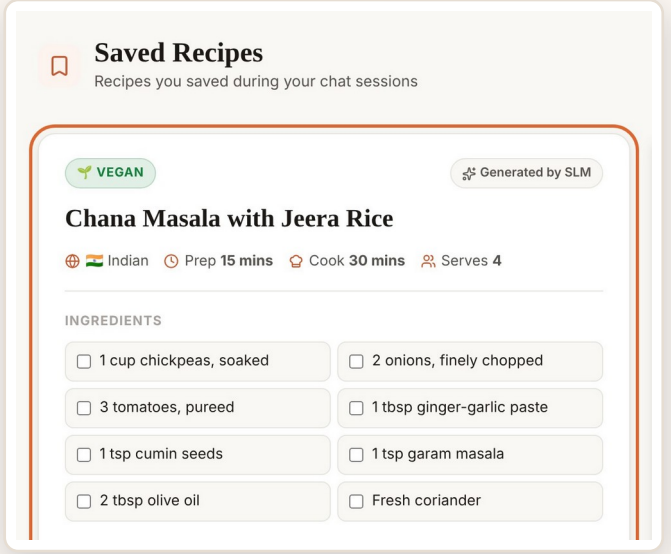


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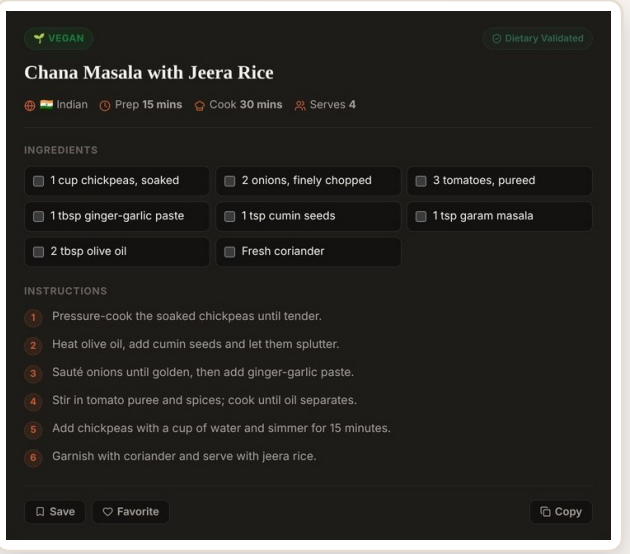
From picking a diet to saving a recipe



1 **Pick your diet once**



2 **Save what you like**



3 **Light or dark theme**

Local Mac

`./run_on_mac.command`

GPU server

`Uvicorn + Nginx, HTTPS`

Docker

`docker compose up`

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Smarter recipes, safe by design

Working product
Recipes, nutrition and 24 diet plans

Fine-tuning works
Loss −62%, complete recipes 97%

Safe by design
Diet and 10 allergens checked

Private and tested
Runs locally · 73 tests pass

Next up

- Hidden ingredients in packaged foods
- Bigger training run
- Faster quantized inference
- More Indian recipes
- Mobile app and reminders

github.com/sreenidhi-km/nourish-slm

References: Qwen3 Technical Report (2025) · Hu et al., LoRA (2022) · Dettmers et al., QLoRA (2023) · Majumder et al., Food.com (2019) · USDA FoodData Central (2018) · Thomas et al., ACSM (2016)

Thank you

Questions are welcome

github.com/sreenidhi-km/nourish-slm

KM Sreenidhi · CSE (AI & ML), Woxsen University

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